Hard AbsTopK gate after a row-sparse linear layer on MNIST ($S = 4$ weights per unit): hard backward against DUST steps on W_0 at four population sizes

2026-10-26

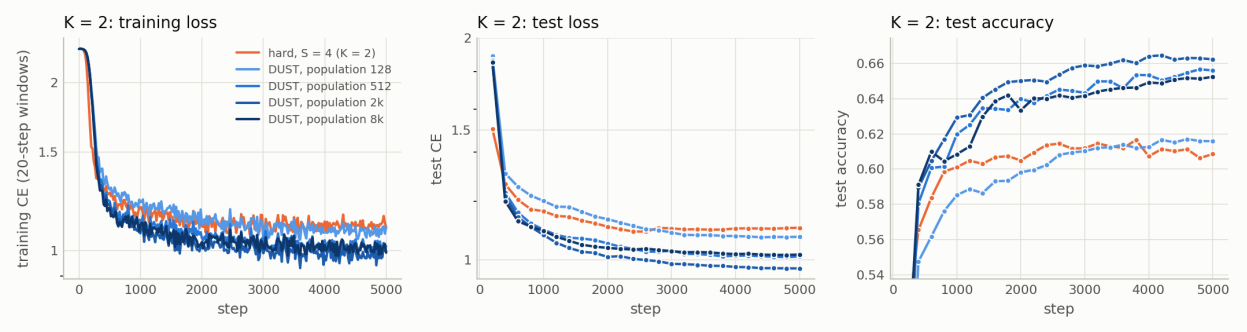


Figure 1: $K = 2$: training CE (left, log scale, 20-step window means), test CE (middle, log scale, every 200 steps) and test accuracy (right) for $z = W_2\,\phi(W_1\,\mathrm{AbsTopK}_K(W_0 x))$ with W_0 (784×784, no bias) projected after every optimizer step and at initialization to its $S = 4$ entries of largest magnitude per row (the incoming weights of a unit), the gradient and the AdamW update of W_0 staying dense. Hard backward alone against every second step a DUST step at population $P \in \{128, 512, 2000, 8000\}$: the gate input $u = W_0 x$ is perturbed on all 784 units with $\mathcal{N}(0, \sigma^2)$ noise, $\sigma = 0.1$, and the estimate $\hat{g}_t$ of $\partial L/\partial u_t$ is applied to W_0 as $\hat{G}_{W_0} = \sum_t \hat{g}_t x_t^\top$ before the projection. Hidden width 1024, 5000 steps (2500 hard steps for W_1, W_2 in the DUST runs against 5000).

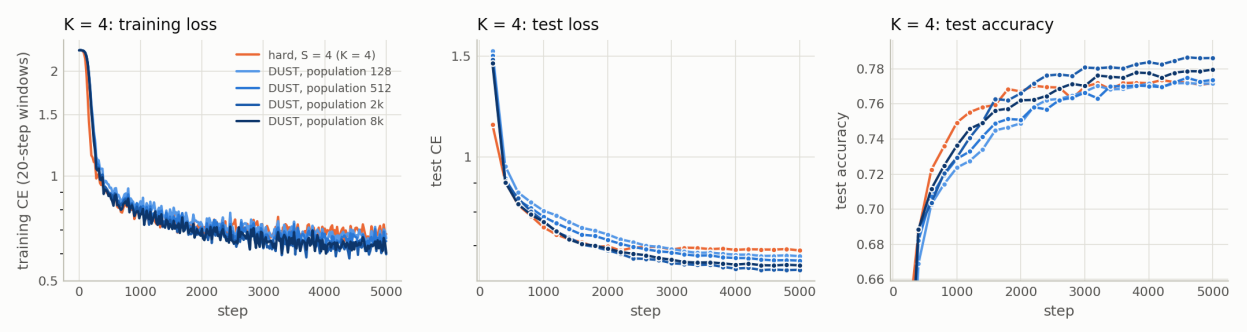


Figure 2: $K = 4$: as Figure 1.

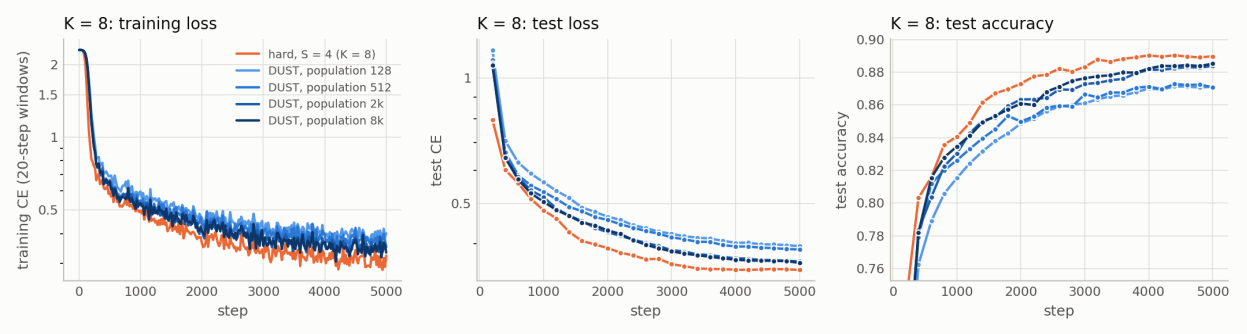


Figure 3: $K = 8$: as Figure 1.

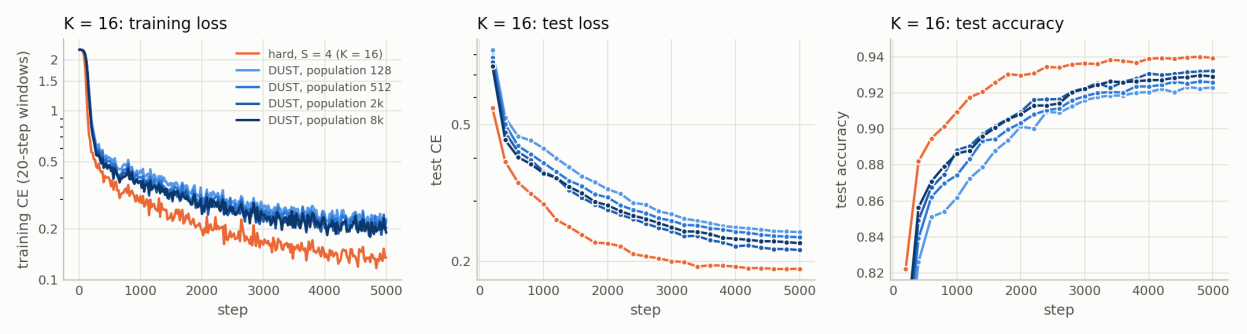


Figure 4: $K = 16$: as Figure 1.